

Forecasting pharmacy demand

What nine models taught us about stock decisions when demand shifts

The simplest model won, but none met the industry accuracy benchmark

9

model families tested on
2,106 days of daily sales

+48%

jump in average demand
between training and test
years

26 to 43%

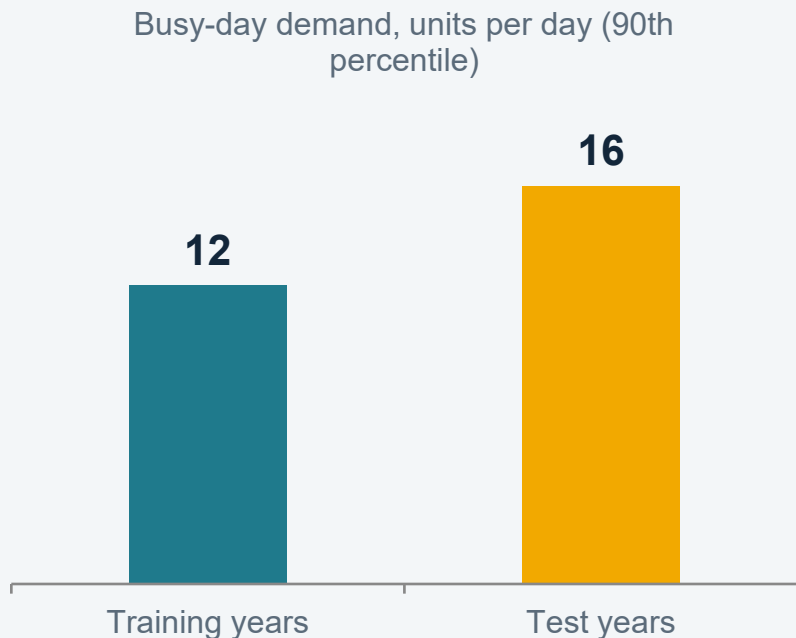
under-prediction in the
leading models, the stock-
out risk

0 of 9

reached the industry error
benchmark of under 35%

So what: demand changed faster than the models could follow. A plain linear model coped best, but every leading model ordered too little. Fix that before anyone relies on the forecast.

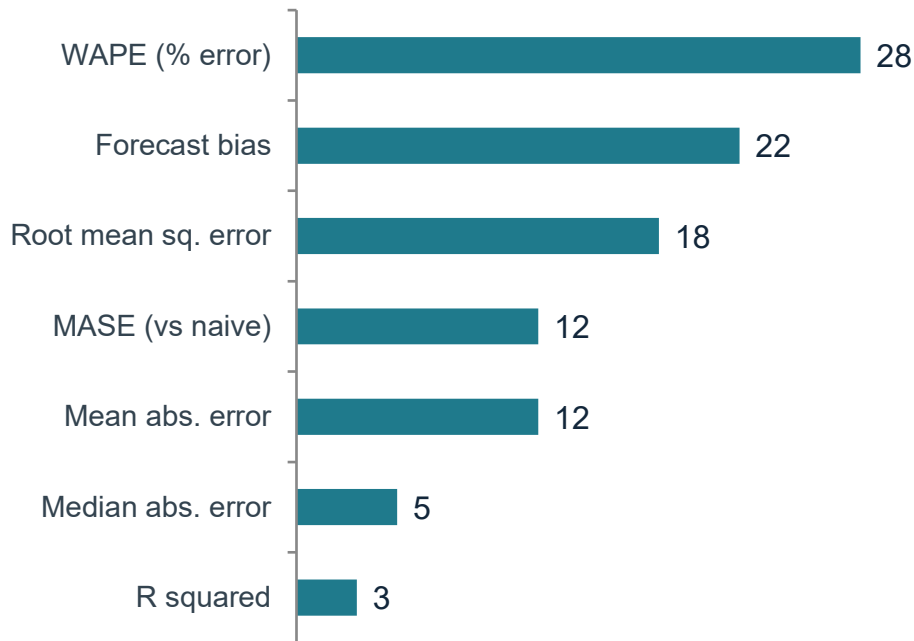
Why it happened: demand in 2018 to 2019 sat above anything the models had seen



What the data showed

- Average demand rose 48% in the test years compared with training.
- Tree-based models cannot predict above the levels they trained on, so they hit a ceiling near 12 units.
- The log-based linear model scales with demand, so it kept pace better.
- Extra tuning did not help, because settings tuned on 2017 did not transfer.

How I judged the models: weighted towards what hurts a pharmacy

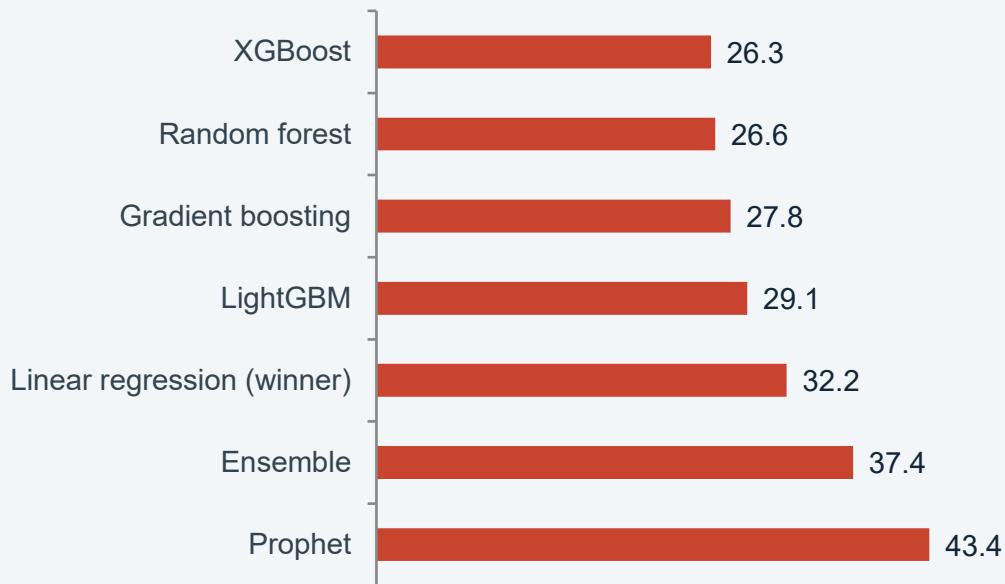


Why not just use accuracy?

- Under-ordering causes stock-outs and patient harm; over-ordering causes waste. A plain average error treats both alike.
- So 62% of the score sits on percentage error, bias and performance against a naive forecast.
- Every candidate was tested in time order, never on shuffled data, to avoid using the future to predict the past.

The catch: every model under-predicted demand, by 26% to 43%

Average under-prediction, % (untuned models, 2018 to 2019)



What it means

- The winner beat the naive forecast by only a small margin (MASE 0.92 vs 1.08).
- Its error was still 65.8% (WAPE), far above the under-35% industry benchmark.
- Ordering from these forecasts would leave shelves short in busy periods.

What I would do next

1 Correct the bias

Apply a bias correction or a loss function that penalises under-prediction more, matching the cost of a stock-out.

2 Watch for drift

Monitor forecast error and retrain when demand shifts, instead of waiting for the yearly cycle.

3 Add outside signals

Flu-season severity, promotions and competitor data were not available and would likely help.

4 Hold off on deployment

No model met the industry benchmark, so use forecasts as decision support only, with human review.